

Al-Usra Hajj and Umrah Company Management System

Database Project Report

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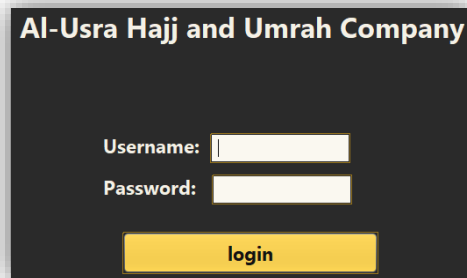


Introduction

The Al-Usra Hajj and Umrah Company Management System is a desktop database application designed to manage the main operations of a Hajj and Umrah company. The system organizes important company data such as users, employees, Hajj applicants, campaigns, packages, bookings, hotels, transportation, tasks, and internal messages in one structured database.

The system was developed using Java Swing for the graphical user interface and MySQL as the relational database. It provides different dashboards for different user roles, including Manager, Employee, and Customer. Each role has its own permissions and available functions. For example, the Manager can manage company records, the Employee can view tasks and messages, and the Customer can view booking, hotel, and transport information.

The main goal of the project is to reduce manual work, improve data organization, and make the company operations easier to manage through a user-friendly application connected to a relational database.



Problem Statement

Hajj and Umrah companies deal with many types of information every day. This information includes customer records, Hajj applicant details, employee data, hotel reservations, transportation details, packages, bookings, tasks, and messages. Managing all this data manually can lead to several problems, such as duplicated records, missing information, slow searching, and difficulty in updating or tracking customer services.

Another important issue is user permissions. In a real company, not all users should access the same information. The Manager needs full access to manage company data, while Employees should only access their tasks, messages, and profile information. Customers should only view their own profile, booking, hotel, and transport details.

For these reasons, the system was designed to provide a centralized database with a clear graphical interface. The system helps organize the company's data, reduce errors, improve access control, and make daily operations easier and faster.

Project Objectives

The main objectives of this project are:

1. To design a database system for managing a Hajj and Umrah company.
2. To store company data in a structured MySQL relational database.
3. To create a Java Swing desktop application connected to the database.
4. To provide a role-based login system for Manager, Employee, and Customer.
5. To allow the Manager to manage campaigns, Hajj applicants, employees, hotels, and transport companies.
6. To allow Employees to view and update their profile information, manage assigned tasks, and read messages.
7. To allow Customers to view their profile, booking, hotel, and transport information.
8. To apply CRUD operations such as adding, viewing, updating, and deleting records.
9. To use primary keys and foreign keys to connect related tables correctly.
10. To make the system organized, easy to use, and suitable for a real company environment.

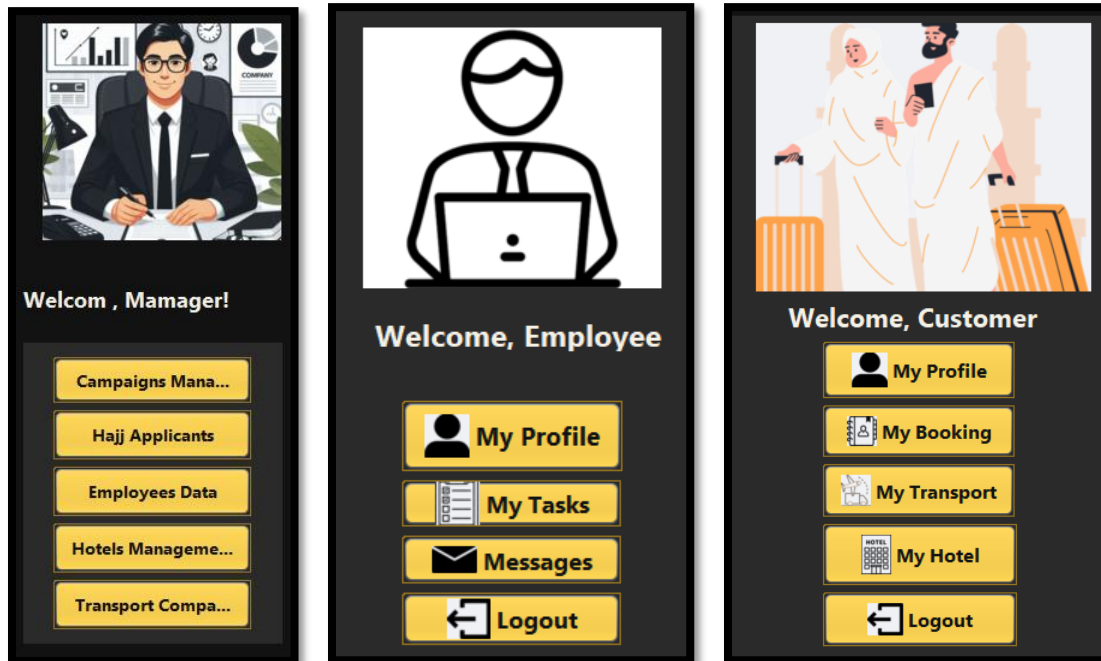
System Users and Roles

The system is divided into three main user roles: Manager, Employee, and Customer. Each role has a different dashboard and different permissions based on the tasks that the user should perform inside the system.

The Manager has the highest level of access. The Manager can manage the main company records, including campaigns, Hajj applicants, employees, hotels, and transport companies. This role is responsible for maintaining and updating the main data of the system.

The Employee has limited access compared to the Manager. The Employee can view personal profile information, check assigned tasks, update task status, and read or reply to messages. The Employee cannot manage all company records.

The Customer can access only customer-related services. The Customer can view personal Hajj information, booking details, hotel reservation, and transport information. The Customer can also edit limited data such as phone number and password.



Database Design and Main Tables

The system uses a MySQL relational database to store and organize all project data. The database is designed as a group of related tables, where each table represents a specific part of the system. For example, the users table stores login accounts, the employees table stores employee information, and the hajj_applicants table stores customer or pilgrim information.

The database design helps reduce data duplication and keeps the information organized. Instead of storing all data in one large table, the system separates the data into smaller related tables. This makes the database easier to manage, update, and understand.

The main database tables include users, hajj_applicants, employees, campaigns, packages, bookings, hotels, hotel_bookings, transport_companies, transports, hajji_transports, employee_tasks, and employee_messages. These tables work together to support the main functions of the Manager, Employee, and Customer dashboards.

Hajj Report

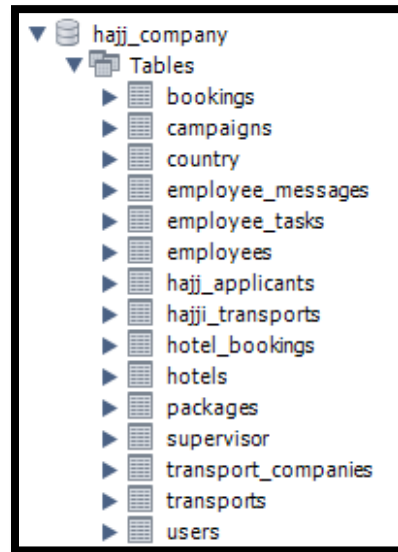


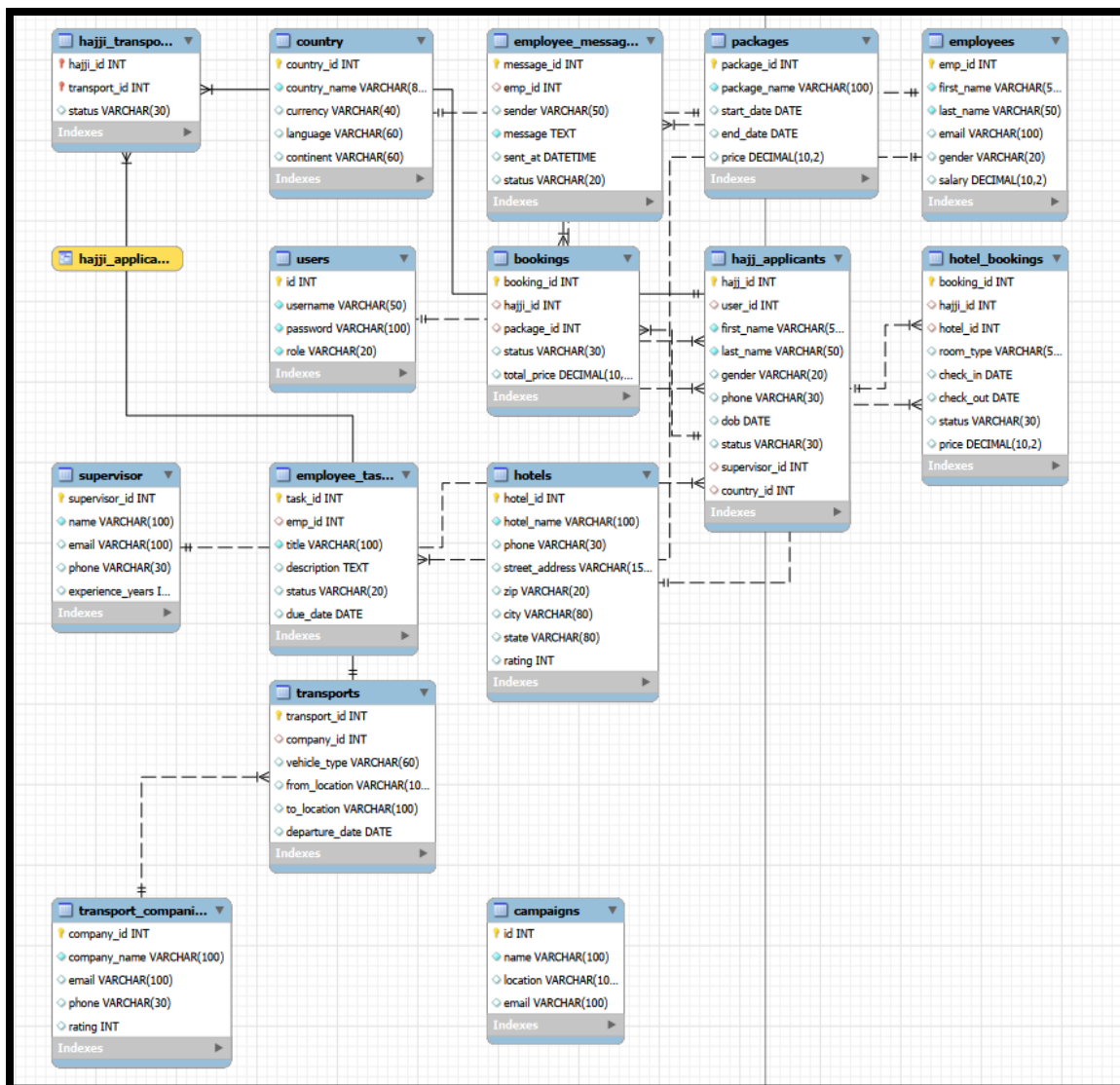
Table Name	Purpose
Users	Stores login information such as username, password, and role.
hajj_applicants	Stores customer or pilgrim information and application status.
Employees	Stores employee information such as name, email, gender, and salary.
Campaigns	Stores Hajj campaign information such as name, location, and email.
Packages	Stores Hajj package details such as package name, start date, end date, and price.
Bookings	Connects a Hajj applicant with a selected package and stores booking status.
Hotels	Stores hotel information such as name, city, phone, address, and rating.
hotel_bookings	Connects a Hajj applicant with a hotel reservation.
transport_companies	Stores transport company information such as name, email, phone, and rating.
Transports	Stores transport trip details such as vehicle type, locations, and departure date.
hajji_transports	Connects Hajj applicants with transport trips.
employee_tasks	Stores tasks assigned to employees.
employee_messages	Stores internal messages between admin and employees.

Relationships Between Tables

The database tables are connected using primary keys and foreign keys. A primary key is used to uniquely identify each record in a table, while a foreign key is used to connect one table with another table.

For example, the `employee_tasks` table contains `emp_id` as a foreign key, which connects each task to a specific employee in the `employee's` table. Also, the `bookings` table contains `hajji_id` and `package_id`, which connect each booking to a specific Hajj applicant and a specific Hajj package.

The hotel_bookings table connects Hajj applicants with hotels, while the hajji_transports table connects Hajj applicants with transport trips. This relational design makes the system organized and prevents storing repeated information in multiple places.



Hajj Report

Relationship	Type	Description
employees → employee_tasks	One-to-Many	One employee can have many assigned tasks.
employees → employee_messages	One-to-Many	One employee can have many messages.
hajj_applicants → bookings	One-to-Many	One Hajj applicant can have booking records.
packages → bookings	One-to-Many	One package can be used in many bookings.
hotels → hotel_bookings	One-to-Many	One hotel can have many hotel bookings.
hajj_applicants → hotel_bookings	One-to-Many	One Hajj applicant can have hotel booking records.
transport_companies → transports	One-to-Many	One transport company can provide many transport trips.
hajj_applicants → hajji_transports	One-to-Many	One Hajj applicant can be assigned to transport trips.
transports → hajji_transports	One-to-Many	One transport trip can be assigned to many Hajj applicants.

Constraints and Data Integrity

The database uses several constraints to protect the data and keep it correct. These constraints help prevent invalid records, duplicated values, and broken relationships between tables.

The system uses primary keys to uniquely identify records. For example, each employee has a unique `emp_id`, each Hajj applicant has a unique `hajj_id`, and each booking has a unique `booking_id`. The database also uses foreign keys to connect related tables together. For example, `bookings.hajji_id` refers to `hajj_applicants.hajj_id`, and `bookings.package_id` refers to `packages.package_id`.

Some tables also use additional constraints such as `NOT NULL`, `UNIQUE`, `CHECK`, and `AUTO_INCREMENT`. The `NOT NULL` constraint prevents important fields from being empty. The `UNIQUE` constraint prevents duplicate values such as usernames. The `CHECK` constraint limits values to a valid range, such as hotel rating from 1 to 5. The `AUTO_INCREMENT` feature automatically generates ID values.

These constraints make the database more reliable and reduce the chance of data entry errors.

```
CREATE TABLE `hotels` (  
  `hotel_id` INT AUTO_INCREMENT NOT NULL,  
  `hotel_name` VARCHAR(100) NOT NULL,  
  `phone` VARCHAR(30),  
  `street_address` VARCHAR(150),  
  `zip` VARCHAR(20),  
  `city` VARCHAR(80),  
  `state` VARCHAR(80),  
  `rating` INT DEFAULT 1,  
  CONSTRAINT `hotels_rating_check` CHECK (`rating` >= 1 AND `rating` <= 5),  
  CONSTRAINT `hotels_pkey` PRIMARY KEY (`hotel_id`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```


SQL Implementation

The SQL script is responsible for building the database structure and inserting the sample data used in the project. It includes commands for creating tables, defining primary keys and foreign keys, adding constraints, and inserting initial records.

The script uses `CREATE TABLE` to create the database tables. Each table contains columns with suitable data types, such as `INT` for numeric IDs, `VARCHAR` for text values, `DATE` for dates, and `DECIMAL` for prices and salaries.

The project also uses SQL commands for daily operations. The `INSERT` command is used to add new records, `SELECT` is used to retrieve data and display it in the interfaces, `UPDATE` is used to modify existing records, and `DELETE` is used to remove records when needed.

For example, when the Manager adds a new employee, the system executes an `INSERT` query. When the Manager updates employee data, the system executes an `UPDATE` query. When the interface loads the table data, the system executes a `SELECT` query.

Main System Interfaces

The system contains several graphical interfaces designed according to the user role. Each dashboard provides different functions depending on the permissions of the logged-in user. The main dashboards are Manager Dashboard, Employee Dashboard, and Customer Dashboard.

The Login interface is the first screen in the system. The user enters a username and password, and the system checks the information from the users table. Based on the user role, the system opens the correct dashboard.

The Manager Dashboard gives access to the main management screens. The Manager can manage campaigns, Hajj applicants, employees, hotels, and transport companies. These screens allow the Manager to add, update, delete, and view records.

The Employee Dashboard allows employees to view their profile, tasks, and messages. Employees can update limited profile information, mark tasks as done, and read or reply to messages.

The Customer Dashboard allows customers to view their Hajj profile, booking information, hotel reservation, and transport details. Customers have limited access and can only view their own related information.

Manager Dashboard Interfaces

The Manager Dashboard is responsible for managing the main company data. It includes several management sections such as Campaigns Management, Hajj Applicants Management, Employees Data Management, Hotel Management, and Transport Company Management.

In the Campaigns Management screen, the Manager can manage Hajj campaigns by adding new campaigns, updating campaign information, deleting campaigns, and viewing existing campaign records. The table displays campaign data such as ID, name, location, and email.

In the Hajj Applicants Management screen, the Manager can manage pilgrims or customers who applied for Hajj services. The screen includes fields such as first name, last name, gender, date of birth, phone number, and status. The status field helps the company track whether the application is pending, approved, rejected, cancelled, or completed.

The Employees Data Management screen allows the Manager to manage employee records and send internal messages. The Manager can add employees, update their information, delete employees, and send messages that can be viewed by employees in their dashboard.

The Hotel Management and Transport Company Management screens allow the Manager to manage hotels and transport companies used by the company. These screens support the same CRUD operations and help organize service providers.

The screenshot shows the 'Campaigns Management' interface. On the left is a sidebar with a user profile and navigation buttons: 'Campaigns Mana...', 'Hajj Applicants', 'Employees Data', 'Hotels Manageme...', and 'Transport Compa...'. The main area has a title 'Campaigns Management' with 'Add', 'Update', 'Delete', and 'Clear' buttons. Below are input fields for 'ID:', 'Name:', 'Location:', and 'Email:'. A 'Data description table' is shown with one row of data.

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com

The screenshot shows the 'Hajjaj Management' interface. It features a similar sidebar to the previous screen. The main area has a title 'Hajjaj Management' with 'Add', 'Update', 'Delete', and 'Clear' buttons. Below are input fields for 'ID:', 'First:', 'Last:', 'Gender:' (with a dropdown), 'DOB:', 'Phone:', and 'Status:' (with a dropdown). A 'Data description table' is shown with one row of data.

ID	First Name	Last Name	Phone	status	Gender	DateOfBirth
1	bb	User	45454545	Pending	Male	2000-01-01

Hajj Report



Welcom , Manager!

- Campaigns Mana...
- Hajj Applicants
- Employees Data
- Hotels Manageme...
- Transport Compa...

Employee data management

Add Update Delete Clear

ID: First: Last: Salary:

Email: Gender: Male

Data description table

ID	First Name	Last Name	Email	Gender	Salary
1	bb	noor	ahmad@example.com	Male	700.00
2	ameen	emad	ameen@gmail.com	male	100.00



Welcom , Manager!

- Campaigns Mana...
- Hajj Applicants
- Employees Data
- Hotels Manageme...
- Transport Compa...

Hotel Management

Add Update Delete Clear

ID: Name: Phone:

City: State: Street: ZIP:

Hotel description table

ID	Hotel Name	Phone	Street Address	ZIP	City	State	Rating
1	Makkah Hotel	123456789	Central Area	24231	Makkah	Makkah Region	5



Welcom , Manager!

- Campaigns Mana...
- Hajj Applicants
- Employees Data
- Hotels Manageme...
- Transport Compa...

Transport Company Management

Add Update Delete Clear

ID: Name: Phone: Rating:

Email:

Hotel description table

ID	Company_name	Email	Phone	Rating
1	Haji Transport Co	transport@example.com	55555555	4

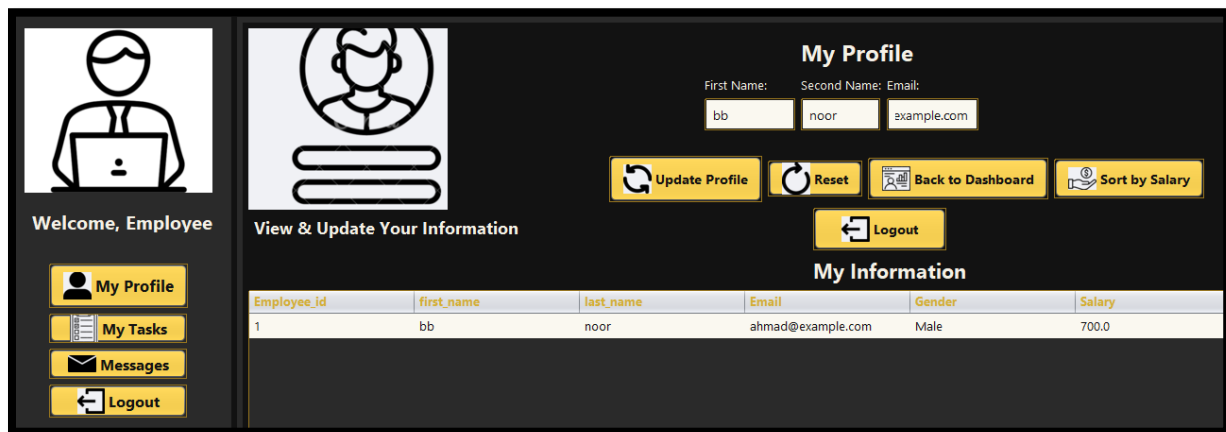
Employee Dashboard Interfaces

The Employee Dashboard is designed for employees who work inside the company. It provides access to three main sections: My Profile, My Tasks, and My Messages.

The My Profile screen displays employee information such as employee ID, first name, last name, email, gender, and salary. The employee can update limited personal information, while sensitive data is controlled by the system or Manager.

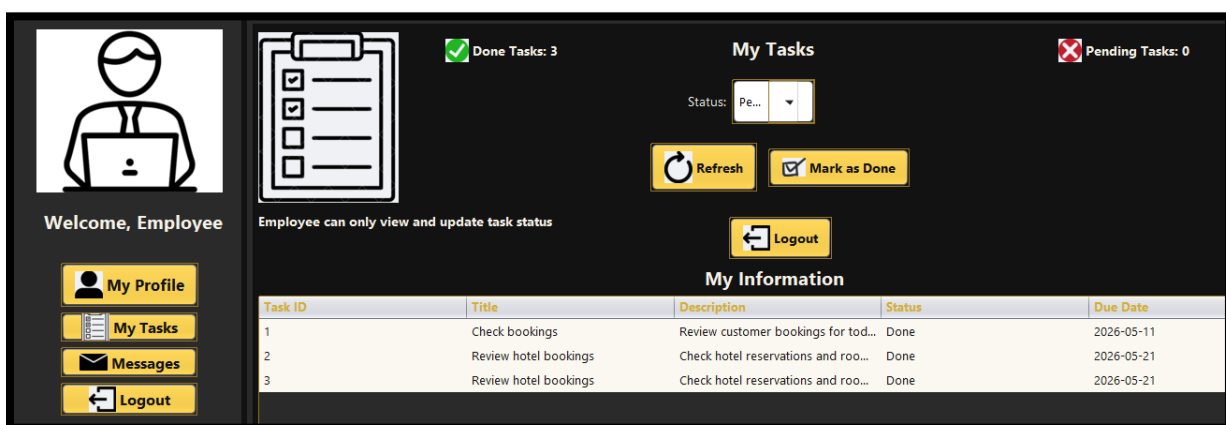
The My Tasks screen displays tasks assigned to the logged-in employee. The employee can view task title, description, status, and due date. The employee can also mark tasks as done. The system shows the number of pending and completed tasks to make task tracking easier.

The My Messages screen displays internal messages between the employee and the administration. The employee can refresh messages, mark messages as read, and reply to messages. This feature improves communication inside the company.



The 'My Profile' dashboard interface features a sidebar on the left with a 'Welcome, Employee' message and navigation buttons for 'My Profile', 'My Tasks', 'Messages', and 'Logout'. The main content area is titled 'My Profile' and includes a user icon, input fields for 'First Name' (bb), 'Second Name' (noor), and 'Email' (example.com). Below these are buttons for 'Update Profile', 'Reset', 'Back to Dashboard', 'Sort by Salary', and 'Logout'. A section titled 'My Information' contains a table with employee details.

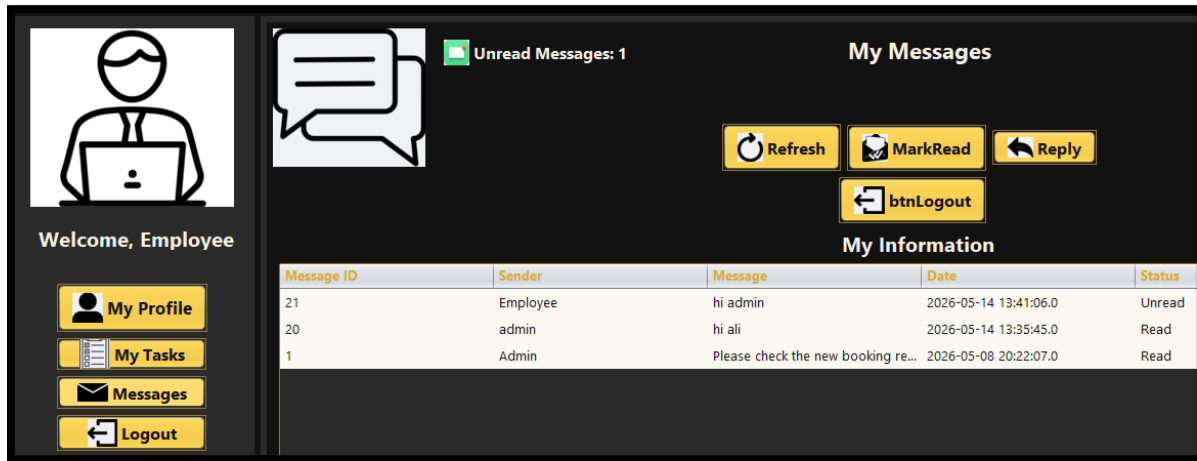
Employee_id	first_name	last_name	Email	Gender	Salary
1	bb	noor	ahmad@example.com	Male	700.0



The 'My Tasks' dashboard interface features a sidebar on the left with a 'Welcome, Employee' message and navigation buttons for 'My Profile', 'My Tasks', 'Messages', and 'Logout'. The main content area is titled 'My Tasks' and includes a clipboard icon, a status filter set to 'Pe...', and buttons for 'Refresh', 'Mark as Done', and 'Logout'. It also displays 'Done Tasks: 3' and 'Pending Tasks: 0'. A section titled 'My Information' contains a table with task details.

Task ID	Title	Description	Status	Due Date
1	Check bookings	Review customer bookings for tod...	Done	2026-05-11
2	Review hotel bookings	Check hotel reservations and roo...	Done	2026-05-21
3	Review hotel bookings	Check hotel reservations and roo...	Done	2026-05-21

Hajj Report



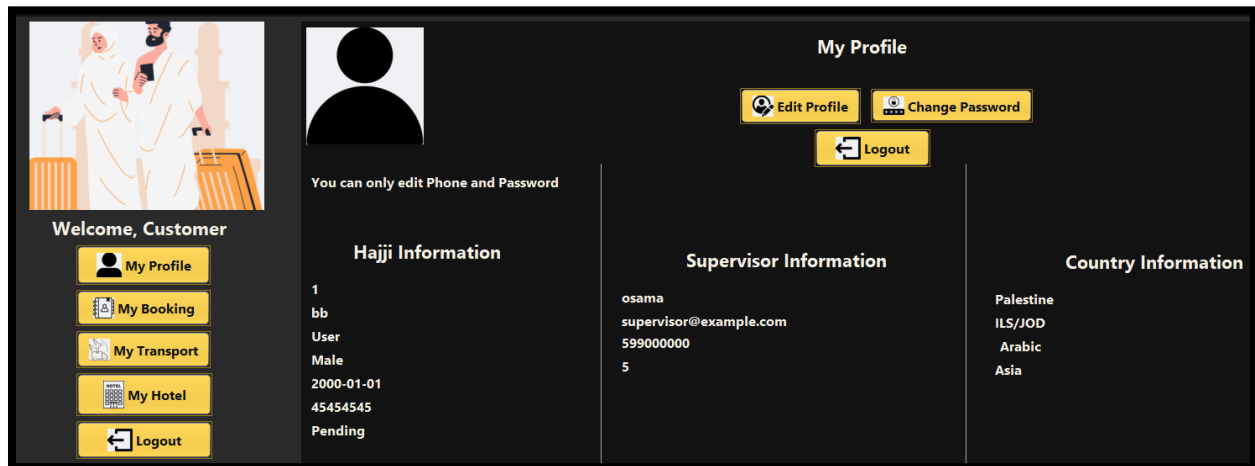
Customer Dashboard Interfaces

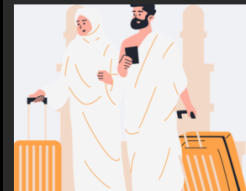
The Customer Dashboard is designed for customers or Hajj applicants. It allows the customer to view personal Hajj information and service-related details.

The My Profile screen displays Hajj applicant information, supervisor information, and country information. The customer can only edit limited information such as phone number and password. This protects important records from unauthorized changes.

The My Booking screen displays booking details such as booking ID, package name, start date, end date, booking status, and total price. The Refresh button reloads the latest booking data from the database, while Cancel Booking updates the booking status.

The My Transport screen displays transportation information such as company name, vehicle type, from location, to location, departure date, and status. The My Hotel screen displays hotel reservation information such as hotel name, city, room type, check-in date, check-out date, status, and price.





Welcome, Customer

My Profile

My Booking

My Transport

My Hotel

Logout



My Booking

Refresh

Cancel Booking

Logout

My Information

Booking ID	Package Name	Start Date	End Date	Status	Total Price
1	Standard Hajj Package	2026-06-01	2026-06-15	cancelled	2500.00



Welcome, Customer

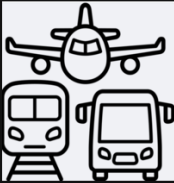
My Profile

My Booking

My Transport

My Hotel

Logout



My Transport

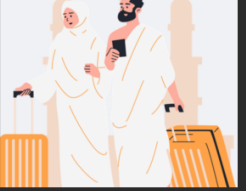
Refresh

Cancel Transport

Logout

My Information

Transport ID	Company Name	Vehicle Type	From Location	To Location	Departure Date	Status
1	Hajj Transport Co	Bus	Jeddah Airport	Makkah	2026-06-01	cancelled



Welcome, Customer

My Profile

My Booking

My Transport

My Hotel

Logout



MyHotel

Refresh

Cancel Hotel

Logout

My Information

Hotel ID	Hotel Name	City	Room Type	Check-In Date	Check-out Date	Status	Price
1	Makkah Hotel	Makkah	Double	2026-06-01	2026-06-10	cancelled	800.00

Hajj Report

Sample Data and Testing

The system was tested using sample data inserted into the MySQL database. The sample data includes users, employees, Hajj applicants, campaigns, packages, bookings, hotels, transport companies, tasks, and messages.

Testing was done by using the graphical interfaces and checking that the database was updated correctly. For example, when a new campaign is added from the Manager Dashboard, the new record appears in the campaigns table. When an employee marks a task as done, the status is updated in the employee_tasks table. When a customer cancels a booking, the booking status changes to cancelled instead of deleting the record completely.

This testing confirms that the application interfaces are correctly connected to the database and that CRUD operations work as expected.

Before :

	hajj_id	user_id	first_name	last_name	gender	phone	dob	status	supervisor_id	country_id
▶	1	3	bb	User	Male	45454545	2000-01-01	Pending	1	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

After :

	hajj_id	user_id	first_name	last_name	gender	phone	dob	status	supervisor_id	country_id
▶	1	3	bb	User	Male	45454545	2000-01-01	Pending	1	1
	2	3	aa	cc	female	056243	2000-01-08	Pending	1	1
*	NULL	NULL	NULL	NULL	CC	NULL	NULL	NULL	NULL	NULL

Normalization

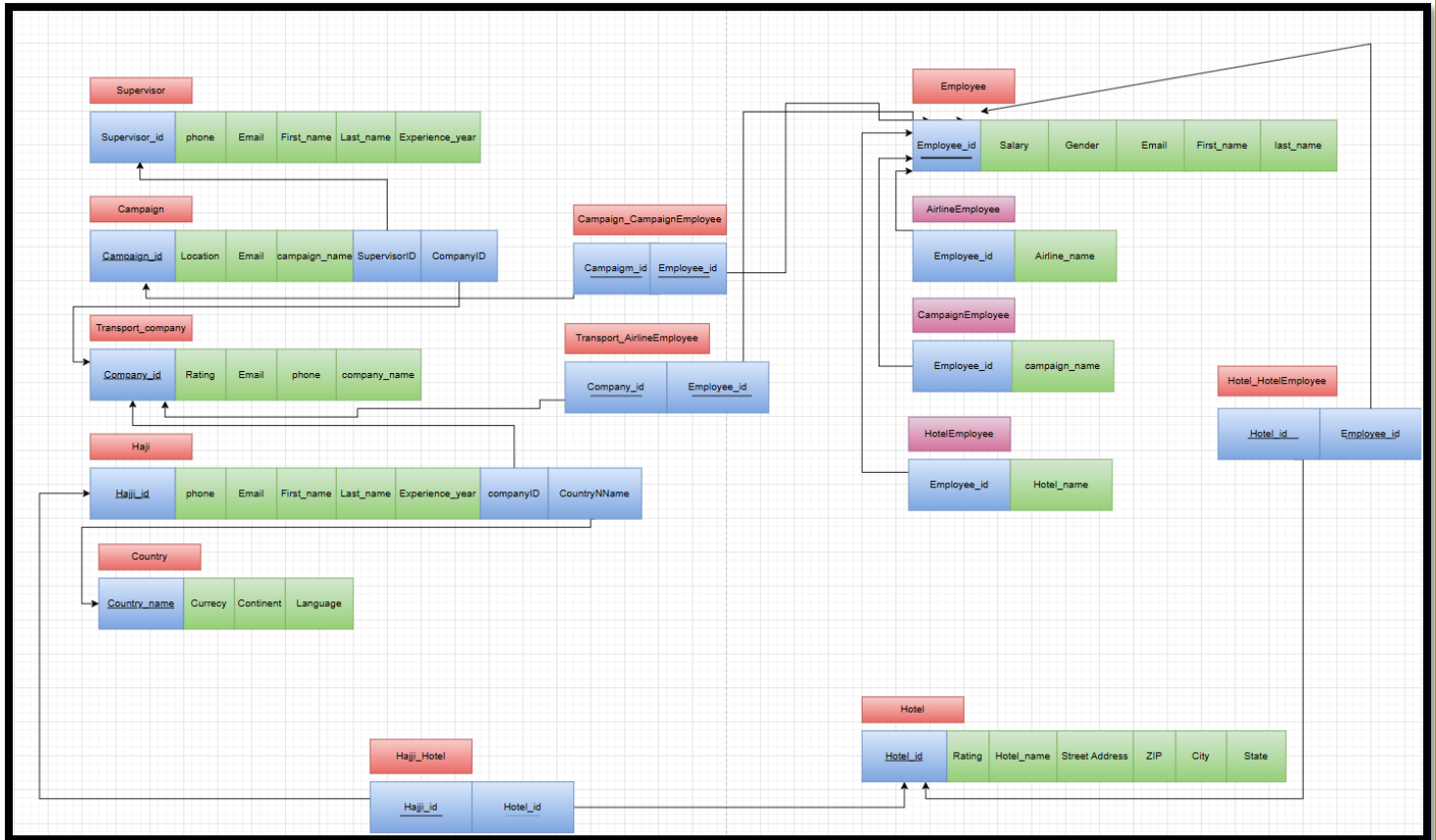
Normalization was applied in the database design to reduce data duplication and improve data organization. Instead of storing all information in one large table, the data was divided into related tables based on the type of information.

The database follows the First Normal Form because each field stores atomic values. For example, employee name, email, gender, and salary are stored in separate columns. It also follows the Second Normal Form because each non-key attribute depends on the primary key of its table. For example, in the employees table, employee name, email, gender, and salary depend on emp_id.

The database also follows the Third Normal Form because related information is separated into different tables. For example, hotel information is stored in the hotels table, while hotel reservation details are stored in the hotel_bookings table. Also, package details are stored in packages, while customer reservations are stored in bookings.

This design reduces repeated data and makes updates easier. For example, if hotel information changes, it can be updated once in the hotels table instead of changing it in many different booking records.

Hajj Report



Security and Access Control

The system uses a role-based login mechanism to control user access. Each user has a username, password, and role stored in the users table. After login, the system checks the role and opens the correct dashboard.

This design improves security because each user can only access the functions related to their role. The Manager can manage company data, Employees can access their tasks and messages, and Customers can view only their personal services such as booking, hotel, and transport information.

The system also uses PreparedStatement in Java to execute SQL queries. This helps make database operations safer and reduces the risk of errors when user input is used in SQL statements.

	id	username	password	role
▶	1	admin	admin123	admin
	2	employee	emp123	employee
	3	customer	cust123	customer
✱	NULL	NULL	NULL	NULL

Conclusion

In conclusion, the Al-Usra Hajj and Umrah Company Management System provides a complete database-based solution for managing the main operations of a Hajj and Umrah company. The system organizes company data using a MySQL relational database and provides a Java Swing graphical interface for different types of users.

The project supports three main roles: Manager, Employee, and Customer. The Manager can manage the main company records, Employees can manage their tasks and messages, and Customers can view their booking, hotel, and transport information. The system applies important database concepts such as primary keys, foreign keys, relationships, constraints, normalization, CRUD operations, and JDBC connectivity.

Overall, the project demonstrates how a real company system can be designed and implemented using database principles and Java programming. It also shows how graphical interfaces can be connected to a database to make data management easier, faster, and more organized.

Tools Used in the Project

In this project, two main programs were used: NetBeans and MySQL Workbench. NetBeans was used as the main development environment for building the Java application. It helped us write and organize the Java code, design the graphical user interfaces using Java Swing, and connect the application screens with the database operations. Through NetBeans, we created the login form, manager dashboard, employee dashboard, and customer dashboard.

MySQL Workbench was used to design and manage the database. It helped us create the database tables, define primary keys and foreign keys, insert sample data, test SQL queries, and view the contents of each table. MySQL Workbench was also useful for exporting the database as a SQL file for submission. Using both tools together allowed us to build a complete system where the Java interfaces in NetBeans interact with the MySQL database created and managed in MySQL Workbench.

